$z \in \mathbb{C}$ $\quad a, b \in \mathbb{R}$

$z = a + ib$

a = parte reale, b = parte immaginaria

$z_1 = 2+3i \quad z_1 \bar{z}_1 = 3-i$

$z_2 = 1-4i \quad z_1 z_2 = 2 - 8i + 3i - 12i^2$

$= \underline{14 - 5i}$

Se moltiplichiamo un numero per se stesso n volte si ha r^n e gli argomenti si sommano n volte, quindi $n\theta$

$$z^n = [r(\cos\theta + i\sin\theta)]^n = r^n(\cos(n\theta) + i\sin(n\theta))$$

$$(z+2i)^{100} = (2\sqrt{2})^{100}(\cos(25\pi) + i\sin(25\pi)) = (2^{3/2})^{100}(-1) = -2^{150}$$

$\cos(\pi) + i\sin(\pi) = -1$, $\sin(\pi) = 0$

$|z| = \sqrt{8} = 2\sqrt{2}$

$z = a + ib$

coniugato $\bar{z} = a - ib$

$|z| = \sqrt{a^2+b^2} \quad \to z \cdot \bar{z} = |z|^2$

inverso $z^{-1} = \frac{1}{z} \quad \to \frac{z_1}{z_2} = \frac{z_1 \cdot \bar{z}_2}{|z_2|^2}$

$\frac{1-i}{2+i} = \frac{(1-i)\cdot(2-i)}{2^2+1} = \frac{2-i-2i+i^2}{5} = \frac{1-3i}{5}$

$\arctan\left(\frac{b}{a}\right) = \arctan(1) = \frac{\pi}{4}$

$\omega = \rho(\cos\varphi + i\sin\varphi) \to \omega^n = z = r(\cos\theta + i\sin\theta)$

$\rho^n(\cos(n\varphi) + i\sin(n\varphi)) = r(\cos\theta + i\sin\theta)$

$\rho^n = r \Rightarrow \rho = \sqrt[n]{r}$

$n\varphi = \theta + 2k\pi \Rightarrow \varphi_k = \frac{\theta + 2k\pi}{n}$ con $k \in \mathbb{Z}$

$$\omega_k = \sqrt[n]{r}\left[\cos\left(\frac{\theta+2k\pi}{n}\right) + i\sin\left(\frac{\theta+2k\pi}{n}\right)\right] \quad \text{per } k = 0, 1, 2, 3, \dots$$

$z = a + ib = r\cos\theta + ir\sin\theta$

$= r(\cos\theta + i\sin\theta)$

$r = |z| = \sqrt{a^2+b^2} \geq 0$

$\to \begin{cases} a = r\cos\theta \\ b = r\sin\theta \end{cases}$

$\theta = \mathrm{Arg}(z)$ (Argomento di z)

Ci servono se ci servono a e b

$\frac{b}{a} = \frac{r\sin\theta}{r\cos\theta} = \tan\theta$

$\theta = \tan^{-1}\left(\frac{b}{a}\right)$

$$\theta = \begin{cases} \arctan\left(\frac{b}{a}\right) & \text{se } a > 0 \quad (\text{I e IV quad.}) \\ \arctan\left(\frac{b}{a}\right) + \pi & \text{se } a < 0 \quad (\text{II e III quad.}) \\ \frac{\pi}{2} & \text{se } a = 0, b > 0 \\ -\frac{\pi}{2} & \text{se } a = 0, b < 0 \end{cases}$$

$z = \sqrt{3} + i = 2\left(\cos\left(\frac{1}{6}\pi\right) + i\sin\left(\frac{1}{6}\pi\right)\right)$

$|z| = \sqrt{3+1} = 2$

$\arctan\left(\frac{1}{\sqrt{3}}\right) = 30° = \frac{1}{6}\pi$ radiani

$z_1 z_2 = r_1 r_2[\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)]$

$z = a + ib \quad \to$ sia $\omega^2 = z$

$\omega = \pm(x + iy)$

$$\omega = \pm\left(\sqrt{\frac{\sqrt{a^2+b^2}+a}{2}} + i\,\mathrm{sgn}(b)\sqrt{\frac{\sqrt{a^2+b^2}-a}{2}}\right)$$

$z = -4i$

$r = 4$

$\theta = \frac{3}{2}\pi$

$z = 4\left(\cos\left(\frac{3}{2}\pi\right) + i\sin\left(\frac{3}{2}\pi\right)\right)$

$\rho = \sqrt[6]{4} = \sqrt[3]{2}$

$\varphi_k = \frac{\frac{3\pi}{2} + 2k\pi}{6} = \frac{\pi}{4} + \frac{k\pi}{3}$ con $\boxed{k = 0,1,2,3,4,5}$

$\omega^2 = \left(\frac{\sqrt{a^2+b^2}+a}{2}\right) + i\left(\frac{\sqrt{a^2+b^2}-a}{2}\right) +$

$+ 2i\sqrt{\frac{(\sqrt{a^2+b^2}+a)(\sqrt{a^2+b^2}-a)}{4}}$

$e^{i\theta} = \cos\theta + i\sin\theta$

Forma esponenziale:

$z = r(\cos\theta + i\sin\theta) \Leftrightarrow z = re^{i\theta}$

- $z_1 \cdot z_2 = (r_1 e^{i\theta_1})(r_2 e^{i\theta_2}) = r_1 r_2 e^{i(\theta_1+\theta_2)}$
- $\frac{z_1}{z_2} = \frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \frac{r_1}{r_2} e^{i(\theta_1-\theta_2)}$
- $z^n = (re^{i\theta})^n = r^n \cdot e^{in\theta}$

$\omega = a + ib$

$e^\omega = e^{a+ib} = e^a \cdot e^{ib} = e^a(\cos b + i\sin b)$

$e^{ix} = \cos x + i\sin x$

$e^{-ix} = \cos(-x) + i\sin(-x) = \cos(x) - i\sin(x)$

$\cos x = \frac{e^{ix} + e^{-ix}}{2}$

$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$

$$\sin(2x) = \frac{e^{2ix} - e^{-2ix}}{2i} = \frac{(e^{ix})^2 - (e^{-ix})^2}{2i} = \frac{(e^{ix} - e^{-ix})(e^{ix} + e^{-ix})}{2i} = \cos x \cdot \frac{(e^{ix} - e^{-ix})}{i} = 2\sin x \cdot \cos x$$

Se un polinomio $p(x) \in \mathbb{R}[x]$ possiede radice complessa $\alpha = a + ib$ (con $b \neq 0$) allora anche il suo complesso coniugato $\bar{\alpha} = a - ib$ deve

$p(x) = \sum_{j=0}^{n} a_j x^j \quad a_j \in \mathbb{R}$

$p(\alpha)=0$ essere rad di $p(x)$

$$\overline{p(\alpha)} = \overline{\sum_{j=0}^{n} a_j \alpha^j} = \sum_{j=0}^{n} \overline{a_j}\,\overline{\alpha^j} = \sum_{j=0}^{n} a_j (\overline{\alpha})^j$$

$\overline{p(\alpha)} = 0$

Trovare le radici di

$$P(x) = x^5 - 2x^4 + 2x^3 - 2x^2 + 4x - 4$$

$x_1 = 1+i$ è una sua radice

Quindi deve essere radice $x_2 = 1-i$ (coniugato)

Quindi se si annulla per $(1+i)$ e $(1-i)$

$$Q(x) = (x-(1+i))(x-(1-i)) = (x-1-i)(x-1+i) = (x^2 - x + ix - x + 1 - i - ix + i - i^2)$$
$$= (x^2 - 2x + 2)$$

$P(x)/Q(x)$

$$\begin{array}{r|l} x^5 - 2x^4 + 2x^3 - 2x^2 + 4x - 4 & x^2 - 2x + 2 \\ \hline -x^5 + 2x^4 - 2x^3 & x^3 - 2 \\ \hline -2x^2 + 4x - 4 & \\ +2x^2 - 4x + 4 & \\ \hline 0 & \end{array}$$

$$P(x) = (x^2 - 2x + 2)(x^3 - 2)$$

$x^3 = 2$ → calcolo di radici complesse

$2 = 2\cdot(1 + 0i) \to \theta = 0$

$2 = 2\cdot(\cos(0) + i\sin(0))$

$k = 0, 1, 2$

$$x_k = \sqrt[3]{2}\left[\cos\left(\frac{2k\pi}{3}\right) + i\sin\left(\frac{2k\pi}{3}\right)\right]$$

Le 5 radici di $P(x)$

$$\{1+i,\ 1-i,\ \sqrt[3]{2},\ x_4,\ x_5\}$$

$k=0 \to x_3 = \sqrt[3]{2}$

$k=1 \to x_4 = \sqrt[3]{2}\left[\cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right)\right] = \sqrt[3]{2}\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$

$k=2 \to x_5 = \sqrt[3]{2}\left(-\frac{1}{2} - i\frac{\sqrt{3}}{2}\right)$

1) a) $\frac{(1-i)}{(2+i)} = \frac{(1-i)(2-i)}{5} = \frac{(2-i-2i+i^2)}{5} = \frac{1}{5} - \frac{3}{5}i$

b) $(2+2i)^{100} = 2^{100}(\cos(100\theta) + i\sin(100\theta))$

$r = |z| = 2\sqrt{2} \qquad \theta = \arctan(1) = \frac{\pi}{4}$

$= 2^{\frac{3}{2}}$

$2^{150}\cdot(\cos(25\pi) + i\sin(25\pi)) =$

$= 2^{150}\cdot(\cos(\pi) + i\sin(\pi)) =$

$= -2^{150}$

c) $(\sqrt{3}+i)^{20}\cdot(1-i)^{10} =$

$(z_1)^{20}\cdot(z_2)^{10} = (z_1^2 z_2)^{10}$

$(\sqrt{3}+i) \to \theta = \arctan(\sqrt{3}) = \frac{\pi}{3}$ | $(1-i) \to \theta = \arctan(1) = \frac{\pi}{4}$

$|z_1| = \sqrt{3+1} = 2$ | $|z_2| = \sqrt{1+1} = \sqrt{2}$

$(2e^{i\frac{\pi}{3}})^2 = 4e^{i\frac{2\pi}{3}}$ | $(\sqrt{2}e^{-i\frac{\pi}{4}})$

$$\left(4\sqrt{2}\,e^{i\left(\frac{2\pi}{3} - \frac{\pi}{4}\right)}\right)^{10} = \left(4\sqrt{2}\,e^{i\frac{5\pi}{12}\pi}\right)^{10}$$

$z_1 = (3+i6)$

$z_2 = (-5+i2)$

$|z_1| = \sqrt{45} = 3\sqrt{5}$

$|z_2| = \sqrt{29}$

$|z_1 z_2| = \sqrt{27^2 + 24^2}$

$|z_1 + z_2| = \sqrt{68} = 2\sqrt{17}$

$z_1 + z_2 = -2 + i8$

$-15 - 6i - 30i - 12i^2$

$\frac{z_1}{z_2} = \frac{z_1\cdot\overline{z_2}}{|z_2|^2} = \frac{(3+i6)(-5-i2)}{29} = -\frac{3}{29} - \frac{39}{29}i$

$\overline{z_1} = (3-i6)$

$\overline{z_1 - z_2} = (3-i6) - (-5-i2) = (8-4i)$

$\overline{z_1 z_2} = (3-i6)(-5-i2) = (-15-6i+30i+12i^2) = (-27+24i)$

$z_1 z_2 = (3+i6)(-5+i2) = (-15+6i-30i+12i^2) = (-27-24i)$

9) a) $x^3 = 1 \to 1(\cos 0 + i\sin 0)$

$k = 0, 1, 2$

$$x_k = \sqrt[3]{1}\left[\cos\left(\frac{2k\pi}{3}\right) + i\sin\left(\frac{2k\pi}{3}\right)\right]$$

$k=0 \to x_1 = 1$

$k=1 \to x_2 = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$

$k=2 \to x_3 = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$

d) $x^8 + x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$

$1(x^2+x+1) + x^3(x^2+x+1) + x^6(x^2+x+1)$

$(x^6+x^3+1)(x^2+x+1)$

(x^6+x^3+1)

$\downarrow y = x^3$

$(y^2+y+1) \qquad y_{1,2} = \frac{-1 \pm \sqrt{-3}}{2} = \frac{-1 \pm i\sqrt{3}}{2}$

$$x_1 = \sqrt[3]{\frac{-1+i\sqrt{3}}{2}} \qquad x_2 = \sqrt[3]{\frac{-1-i\sqrt{3}}{2}}$$

4. Coordinate Polari ed esponenziale

$\theta = \arctan(1) = \frac{\pi}{4}$

$5 = 5(\cos 0 + i\sin 0) = 5e^{i0}$; $-2 = 2(\cos(\pi) + i\sin(\pi)) = 2e^{i\pi}$; $3i = 3\left(\cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right)\right) = 3e^{i\frac{\pi}{2}}$; $-\frac{1}{2}i = \frac{1}{2}\left(\cos\left(\frac{3\pi}{2}\right) + i\sin\left(\frac{3\pi}{2}\right)\right) = \frac{1}{2}e^{i\frac{3\pi}{2}}$; $2+2i = 2\sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) = 2\sqrt{2}e^{i\frac{\pi}{4}}$

$\sqrt{3}+i = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right) = 2e^{i\frac{\pi}{6}}$; $-\sqrt{2}+\sqrt{2}i = 2\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right) = 2e^{i\frac{3\pi}{4}}$; $3+4i = 5\left(\cos(\arctan(\frac{4}{3})) + i\sin(\arctan(\frac{4}{3}))\right) = 5e^{i\arctan(\frac{4}{3})}$

$|z| = \sqrt{3+1} = 2$

$\theta = \arctan(\sqrt{3}) = \frac{\pi}{3}$ $\quad |z| = 2$ $\quad |z| = \sqrt{9+16} = 5$, $\arctan\left(\frac{4}{3}\right)$

$\frac{b}{a} = -\frac{2\sqrt{3}}{2}$

$\omega = 2 - 2\sqrt{3}i$ $\quad \theta = \arctan(-\sqrt{3}) = -\frac{\pi}{3}$

$|z| = \sqrt{4+4\cdot 3} = \sqrt{4+12} = \sqrt{16} = 4$ $\quad \boxed{4e^{i\frac{5\pi}{3}}}$

-1 $\quad +i\sin(\pi)$
$\cos(\pi)$

$z^6 = \omega^6 = z^{12}e^{i15\pi} = z^{12}(\cos(15\pi) + i\sin(15\pi)) = -z^{12}$

$z^3 = \omega$ $\quad \boxed{\text{radice 3}}$

$z = \sqrt[3]{\omega}$

$$\sqrt[3]{4}\left[\cos\left(\frac{-\frac{\pi}{3}+2k\pi}{3}\right) + i\sin\left(\frac{-\frac{\pi}{3}+2k\pi}{3}\right)\right]$$

$k=0 \rightarrow x_1 = \sqrt[3]{4}\left(\frac{\sqrt{3}}{2} - i\frac{1}{2}\right)$

$k=1 \rightarrow \sqrt[3]{4}\left[\cos\left(\frac{5}{6}\pi\right) + i\sin\left(\frac{5}{6}\pi\right)\right]$

$x_2 = \sqrt[3]{4}\left(+\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$

$k=2 \rightarrow \sqrt[3]{4}\left[\cos\left(\frac{11}{6}\pi\right) + i\sin\left(\frac{11}{6}\pi\right)\right]$

$x_3 = \sqrt[3]{4}\left(\frac{\sqrt{3}}{2} - i\frac{1}{2}\right)$